ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE D · MARKING GUIDELINE

MARKS: 80

Marking guideline for the Academic Central NC(V) Level 4 Electronics and Digital Concepts for Robotics paper 2 · practice d.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Drawings and breadboard layouts may look different and still be correct: check every component, symbol, label and connection. Deduct 1 mark for each wrong connection and 2 marks for any short circuit (no less than 0 for a question).

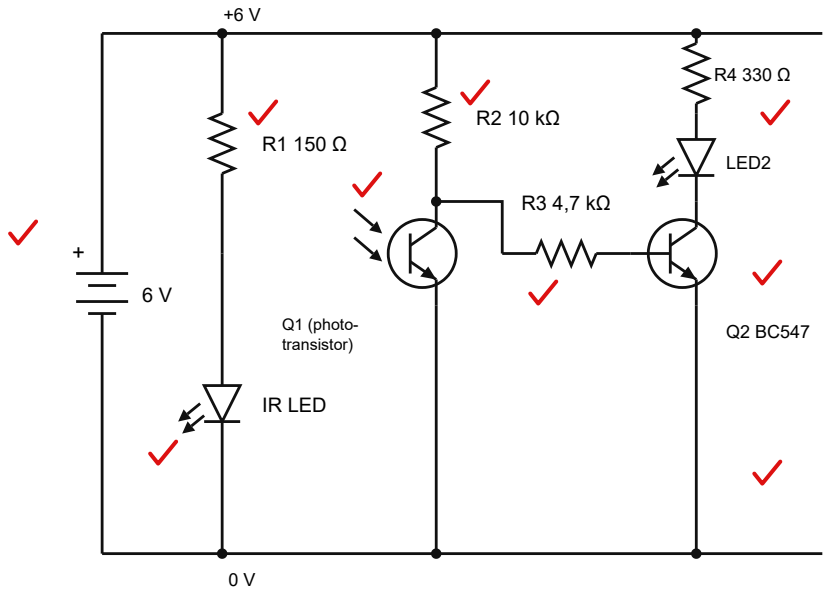
QUESTION 1

+12 V DC input and 0 V line ✓; D1 in series with the + supply, anode towards the input ✓; C1 across the input after D1 with correct polarity ✓; LM317 with IN, OUT and ADJ correctly connected ✓✓; R1 from OUT to ADJ ✓; VR1 (variable resistor) from ADJ to 0 V ✓; C2 across the output with correct polarity ✓; R2 and LED1 across the output with correct LED polarity ✓; all labels and values shown ✓ (10)

[10]

QUESTION 2

Correct circuit: 6 V battery; 150 Ω resistor in series with the IR LED from + to −; 10 kΩ resistor from + to the phototransistor collector, phototransistor emitter to −; 4,7 kΩ resistor from the phototransistor collector to the base of the BC547; BC547 emitter to −; 330 Ω resistor from + to the anode of LED2; LED2 cathode to the BC547 collector.

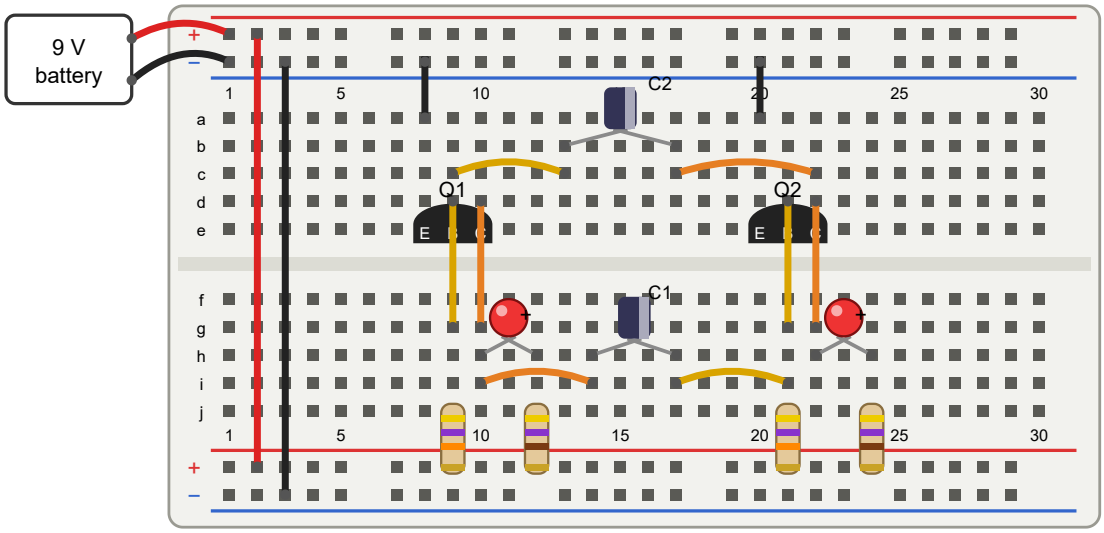


6 V battery symbol with polarity ✓; 150 Ω resistor and IR LED in series, correct polarity ✓; 10 kΩ pull-up to the phototransistor collector ✓; phototransistor symbol, emitter to 0 V ✓; 4,7 kΩ base resistor from the phototransistor collector to the BC547 base ✓; NPN transistor symbol with collector, base and emitter correct ✓; emitter to 0 V ✓; 330 Ω resistor and LED2 from + to the collector ✓; LED2 polarity correct ✓; labels and values ✓ (10)

[10]

QUESTION 3

3.1 +9 V and 0 V lines ✓; R1 and LED1 in series to the collector of Q1 ✓, R4 and LED2 to the collector of Q2 ✓, correct LED polarity ✓; R3 from +9 V to the base of Q1 ✓; R2 from +9 V to the base of Q2 ✓; C1 from the collector of Q1 to the base of Q2 ✓ with + to the collector ✓; C2 from the collector of Q2 to the base of Q1 ✓ with + to the collector ✓; both emitters to 0 V ✓; NPN transistor symbols correct ✓; labels and values ✓✓; crossing lines drawn without junction dots ✓ (15)



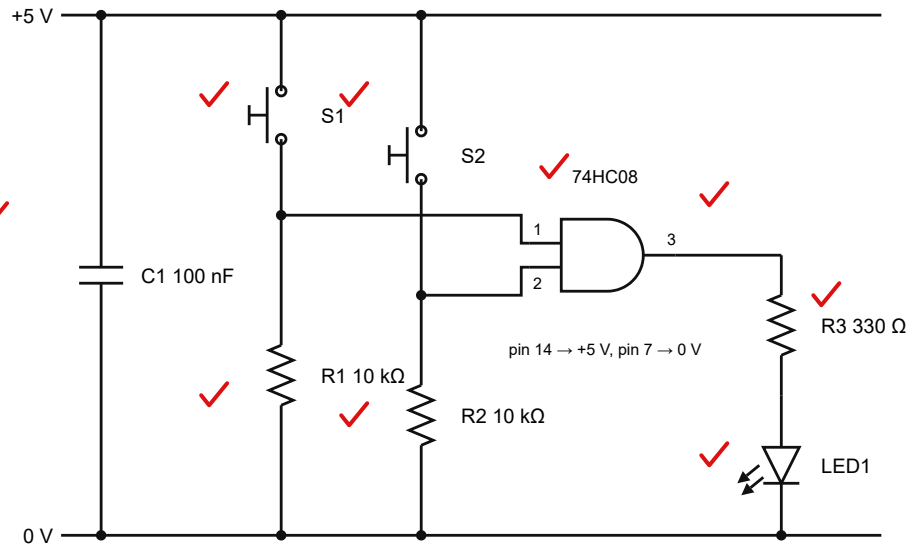
Example of a correct layout for 3.2 (other neat, correct answers get full marks)

3.2 Placement ✓✓✓✓✓✓ (Q1, Q2, R1–R4, C1, C2, LED1, LED2 – any six correctly placed) = 6. Polarity / orientation ✓✓✓: transistor pins C-B-E; C1 and C2; both LEDs = 3. Connections ✓✓✓✓✓✓: R1–LED1 to the Q1 collector and R4–LED2 to the Q2 collector; R3 to the Q1 base and R2 to the Q2 base; C1 cross-coupling Q1 collector to Q2 base; C2 cross-coupling Q2 collector to Q1 base; emitters to 0 V; battery to the rails = 6. (15)

[30]

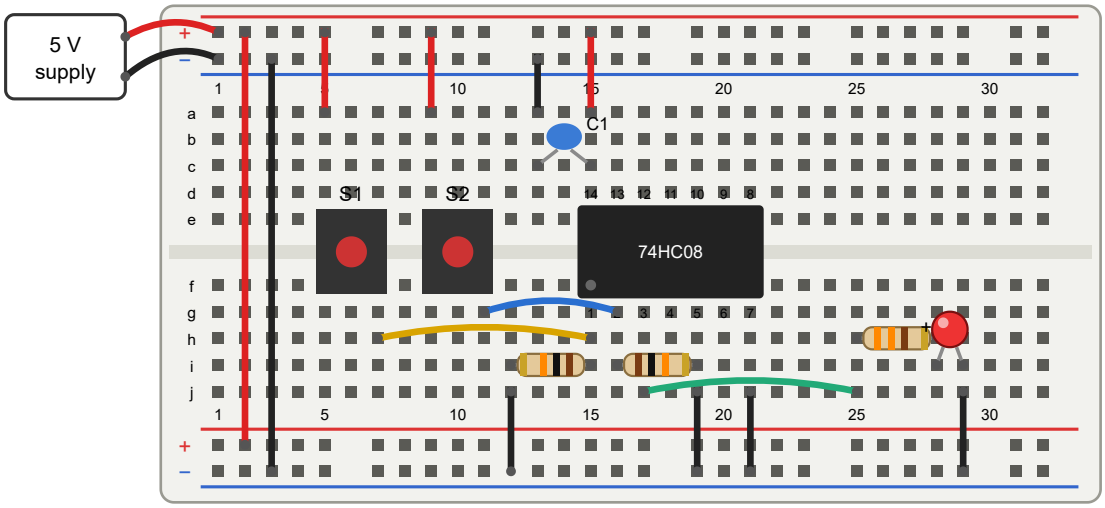
QUESTION 4

4.1



+5 V and 0 V lines ✓; S1 from +5 V to pin 1 ✓ and S2 from +5 V to pin 2 ✓; R1 pull-down from pin 1 to 0 V ✓ and R2 pull-down from pin 2 to 0 V ✓; 74HC08 drawn as a block (or AND gate) with pin numbers ✓✓; pin 14 to +5 V ✓ and pin 7 to 0 V ✓; pin 3 through R3 to LED1 ✓ and LED1 to 0 V with correct polarity ✓; C1 100 nF across the supply ✓; labels and values ✓✓; crossing lines without dots ✓

(15)



Example of a correct layout for 4.2 (other neat, correct answers get full marks)

4.2 Placement ✓✓✓✓✓✓ (74HC08 across the centre gap, S1, S2, R1, R2, R3, LED, C1 – any six correctly placed) = 6. Polarity / orientation ✓✓✓: IC notch / pin 1; LED; battery or supply = 3. Connections ✓✓✓✓✓✓: pin 14 to +5 V and pin 7 to 0 V; S1 and R1 at pin 1; S2 and R2 at pin 2; pull-downs to 0 V; pin 3 to R3 and the LED; C1 across the supply rails = 6. (15)

[30]

TOTAL: 80

END OF MARKING GUIDELINE